

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU’s 8-domain, 41-subject taxonomy is explicitly structured around Indian academic and competitive examination content [Q1, Q17, Q19], with ‘local knowledge’ operationalized as Indian state-level [Q12]. The deployment requires Bangladesh-specific categories (1971 Liberation War, SSC/HSC exams, Bangladeshi universities, Ekushe Boi Mela, rural/riverine contexts) which have no designated position in the ontology. Empirical sampling confirms zero Bangladeshi cultural touchstones across 21 Bengali validation items [DATASET-D2, D3, D4, D5, D6, D7] and that Bangladesh appears only as a distractor option in non-Bengali splits [DATASET-D22, D23]. The authors themselves note models perform poorly in culturally relevant areas [Q6, Q68], precisely the domains where Bangladeshi knowledge should appear.

ID	Evidence
Q1	“MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge.” (p.1)
Q6	“Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM.” (p.1)
Q12	“We focus on region-specific exams to authentically capture local knowledge in the respective language.” (p.2)
Q17	“MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge.” (p.3)
Q19	“MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge.” (p.3)
Q68	“Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields.” (p.7)
D2	ফবেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নরিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি ক-োন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নরিবাচন কমশিনারকে অপসারণ করা যতে পারে ... পুরধান নরিবাচন কমশিনারের সুপারশি রাষ্ট্রপতি — Removal of India’s Election Commissioner — refers to Indian constitutional provisions
D4	‘মঙ্গল ভারত’ কাজটি নমিনলখিতি ক-োন জাতীয় নতোর দ্বারা রচতি হয়ছেলি ... আচার্য বনি-োবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফলিমফয়ের অ্যাওয়ার্ডস, ২০২০-এ সরো সনিমোট-োগ্রাফরি জন্য পুরস্কারটি কে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দলিলতিে কুতুব মনিারের নরিমাণ সম্পন্ন করছেলিনে ... ইলতুতমশি — Qutub Minar, Delhi — India-specific historical monument
D7	বাহামান সাম্রাজ্যের পুরথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D22	ডসিমের 2019 ৯৬ ভারতীয় ঔষধাকোষ... বাঙালিদের — Indian Pharmacopoeia — Bangladesh mentioned as a distractor option, not as subject matter
D23	డిసింబర్ 2019లో భారతీయ ఔషధకోషియా ను గుర్తించిన మూడటి దేశం ఏదీ? ... ఆఫ్ఘనిస్తాన్ — India Pharmacopoeia question — Bangladesh listed as distractor option2

Strengths

- Broad multi-domain coverage spans 8 domains including Arts & Humanities, Law & Governance, and STEM [Q1, Q17, DATASET-D2, D4], which means the structural slots exist even if filled with Indian content — a Bangladesh-targeted

re-population of those slots is conceivable.

- Subject-level granularity (41 subjects) and language-stratified evaluation tables [Q99, Q128] provide a fine-grained reporting infrastructure that could in principle host Bangladesh-specific subjects.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q17	"MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge." (p.3)
Q99	"Table 9: Detailed subject level statistics of MILU across different languages." (p.20)
Q128	"Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages." (p.52)
D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নির্বাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কনো রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D4	'মঙ্গল ভারত' কাজটি নমিনলখিতি কনো জাতীয় নতোর দ্বারা রচিত হযছেলি ... আচার্য বনোবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required Bangladesh-specific categories include the 1971 Liberation War, 1952 Language Movement, SSC/HSC exam content, Bangladeshi universities (Dhaka University, BUET, IUT, BRAC), Ekushe Boi Mela and Pohela Boishakh, post-1971 political history including the 2024 July Uprising, rural/riverine contexts, and Bangladeshi literary figures (Humayun Ahmed, Selim Al Deen) [WEB-3, WEB-5, WEB-7]. Story generation also requires categories for narrative settings (rural village life, river delta, Chittagong Hill Tracts, Dhaka neighborhoods) absent from competitive-exam ontology.

IO-2: Yes — the taxonomy is structurally India-first [Q9, Q12, Q19] and excludes Bangladeshi national categories. Empirically confirmed: zero Bangladeshi cultural items in the 21-example Bengali validation sample [DATASET-D2 through D7].

IO-3: Yes — the taxonomy includes Indian-specific categories (Indian state elections, Indian constitutional law, Filmfare Awards, Indian freedom fighters, Indian state welfare schemes) [DATASET-D2, D3, D5, D9, D10, D11, D12] that are construct-irrelevant or actively misleading for a Bangladeshi cultural-grounding evaluation.

IO-4: Major content-validity gap: the taxonomy under-represents Bangladeshi national knowledge and over-represents Indian state/national knowledge. Authors acknowledge poor model performance in culturally relevant areas [Q6, Q68] but attribute it to training data, not to the benchmark's own India-first ontology [Q9].

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q68	"Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)

D2	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নব্বিবাচনে, আম আদমি পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছেলি? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
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D5	৬৫তম ফলিমফয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফরি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D7	বাহামানি সাম্রাজ্যের প্রথম রাজধানী কী ছিলি? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D9	Which state has topped the India Innovation Index - 2019 published by Niti Aayog? ... Karnataka — India Innovation Index — Indian national governance content
D10	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? — Maharashtra state legislature — India-specific governance at state level
D11	Which state became the first Indian state to launch pension for single women at Rs.1,000/- per month? ... Telangana — Telangana state welfare scheme — Indian state government
D12	Founded in 1950, one of the industrial units owned by Indian Railways is named after the Indian freedom fighter ... Chittaranjan Das — Indian Railways industrial unit named after Indian independence figure
WEB-3	University student literacy supports text-based deployment (https://en.wikipedia.org/wiki/Education_in_Bangladesh)
WEB-5	July 2024 Uprising is a foundational live cultural reference for Bangladeshi students, post-benchmark (https://en.wikipedia.org/wiki/July_Revolution_(Bangladesh))
WEB-7	https://www.chathamhouse.org/2025/11/sheikh-hasina-verdict-and-bangladeshs-upcoming-referendum-signal-transitional-moment-south
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)

Information Gaps

- Whether the larger Bengali test split (6,637 items) contains any organically-sourced Bengali-medium items with Bangladeshi-relevant content (validation sample shows 0%)

Requires Expert Verification

- Native Bangladeshi review of any candidate Bengali items that survive translation-filter to confirm absence of Bangladeshi cultural anchoring

Remediation

Gap 1: Taxonomy lacks Bangladeshi-specific categories (1971 Liberation War, SSC/HSC, Bangladeshi universities, cultural events, rural/riverine contexts, 2024 July Uprising)

Recommendation: Supplement with BLUCK (Bangladesh-sourced cultural/historical MCQ across 23 categories [WEB-15]) and BnMMLU (Bangladeshi NCTB-textbook-sourced MMLU [WEB-18]) to populate the missing ontology slots; do not rely on MILU's Bengali split for cultural-knowledge claims.

ID	Evidence
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-18	BnMMLU sources directly from Bangladeshi NCTB textbooks — viable alternative content provenance (https://arxiv.org/abs/2505.18951)